

Anjali Viramgama

Why AI Models Hallucinate



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First, What Is Hallucination?

AI hallucination means the model gives an answer that sounds correct...

But is actually false.

The scary part?

It often says it with full confidence.

No hesitation.

No warning.

No "I'm not sure."

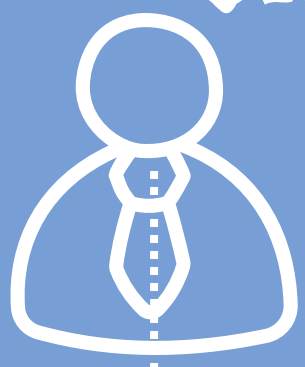
Just a polished wrong answer that feels believable.

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The Simple Example

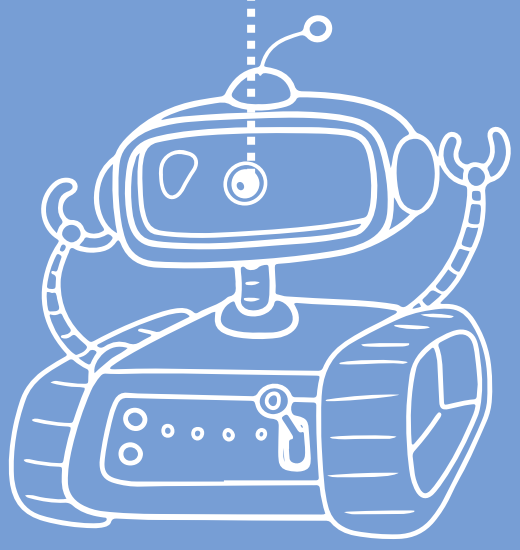


Researchers asked an AI model:
"What is Adam Kalai's birthday?"
A human might say:
"I don't know."
But the AI gave different answers across attempts.

March 7.
June 15.
January 1.

All wrong.

That's hallucination in action.



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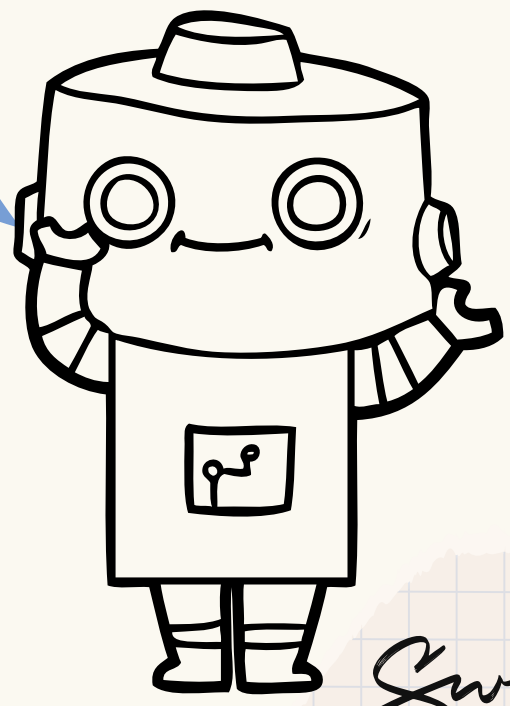
Why Didn't It Say "I Don't Know"?

Because AI models are trained to predict answers.
Not always to admit uncertainty.
If a fact is common online, the model sees it many times.

Example:

Einstein's birthday.
That information appears everywhere.
So the model is more likely to answer correctly.
But if a fact is rare, private, or barely mentioned online,
the model has weak evidence.

So it guesses.



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AI Doesn't "Know" Like Humans

AI models don't store knowledge like a database.

They learn patterns from huge amounts of text.

When you ask a question, the model predicts the most likely next words

So when the answer is not clearly present in its training data, it may still generate something that sounds likely

The answer may sound real, even when it is invented.

That is why rare facts are dangerous.

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Common Facts vs Rare Facts

For example:

- Famous birthdays
- Country capitals
- Historical dates
- Popular company founders
- Well-known public information

These are easier for AI to answer.

For example:

- Unknown birthdays
- Small personal details
- Private company data
- New research updates
- Niche technical information

The less repeated the fact is, the higher the risk of hallucination

Some facts are repeated thousands of times online.

But rare facts are harder

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The Real Problem Is Incentives

Here's where it gets interesting

AI models don't hallucinate only because they lack information.

They hallucinate because the system rewards answering.

Correct answer = points.
Wrong answer = zero.
"I don't know" = often treated like zero too.
So the model learns:
Guessing is better than staying silent.

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The Exam Analogy

Imagine a student taking a test.

If a wrong answer gives zero...

And a blank answer also gives zero...

What will the student do?

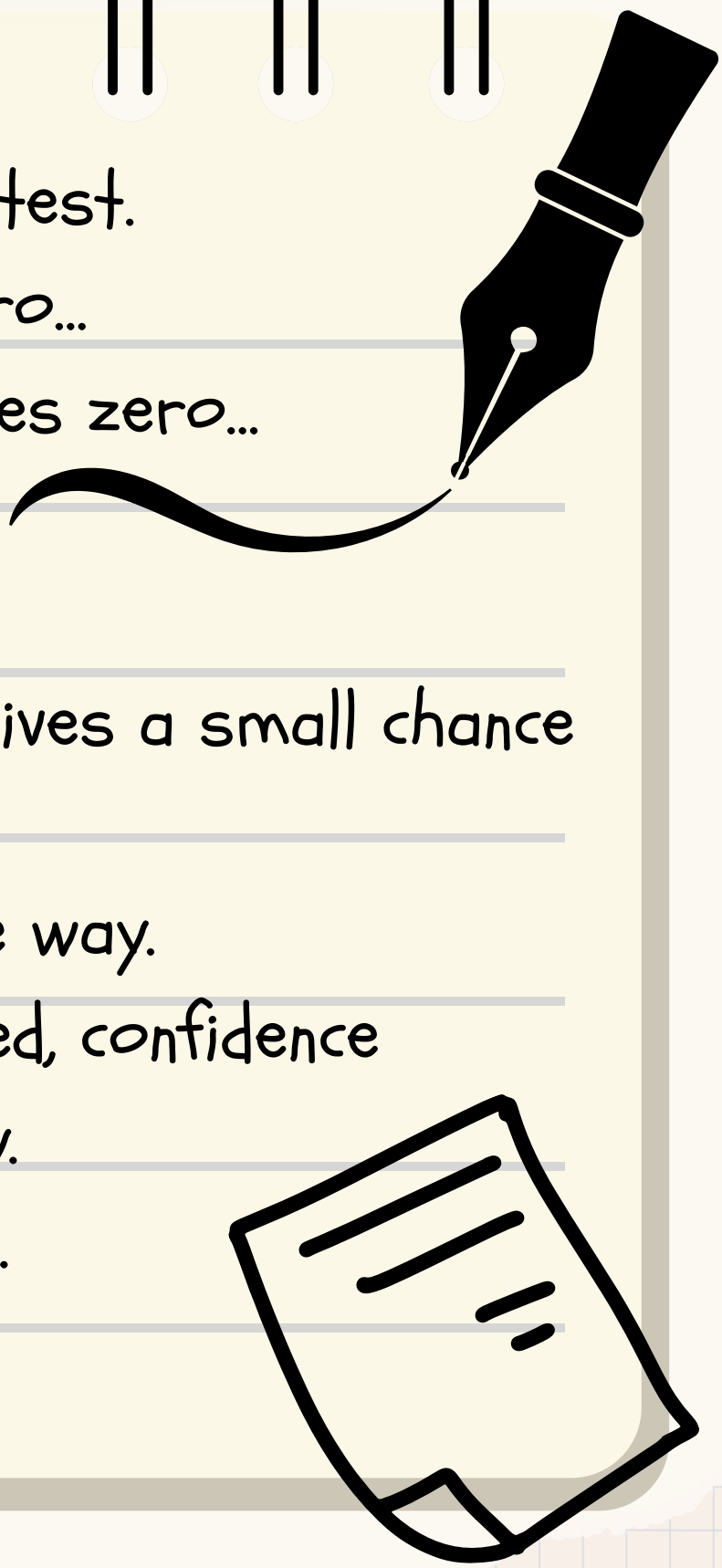
Guess.

Because guessing at least gives a small chance of getting points.

AI models behave the same way.

When uncertainty is punished, confidence becomes the safer strategy.

Even if the answer is wrong.



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We Trained AI to Sound Certain

This is the uncomfortable part.

AI models are often optimized to sound helpful

Helpful usually means:

- Answer quickly
- Be confident
- Avoid refusing

- Complete the task
- Give the user something useful

But honesty sometimes requires saying:
"I don't know."
"I need more context."
"I cannot verify this."
"I may be wrong."
Many systems still don't reward that enough.

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Hallucination Is a System Design Problem

This is not only a model problem.

It is also an evaluation problem.
If AI systems are tested using benchmarks that reward confident guessing, models will keep learning to guess.
So the fix is not just:
"Build a smarter model."
The fix is also:
"Change what we reward."
Because models follow incentives.
Just like people do.

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What Should We Reward Instead?

Better AI evaluation should reward honest uncertainty.
That means models should get credit for saying:
"I don't know."
"This is not available in my data."
"I need a source to verify this."
"This answer is uncertain."
"I can make an estimate, but it may be wrong."
The goal should not be perfect confidence.
The goal should be calibrated confidence.



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Why This Matters for Users

AI hallucinations are not always harmless.
A wrong movie fact is annoying.
But a wrong answer in healthcare, finance, law, hiring, coding, or cybersecurity can create real damage.
That's why users should treat AI as a powerful assistant.
Not a final authority

For important topics:



Ask for sources



Verify facts



Check dates



Compare answers



Avoid trusting confident claims blindly

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How to Reduce Hallucinations

You can reduce hallucinations by prompting better.

Instead of asking:

"Give me the answer."

Ask:

"Answer only if you are confident. If not, say you don't know."

Or:

"Cite sources and separate verified facts from assumptions."

Or:

"Tell me what you are uncertain about."

This pushes the model toward honesty.

Not just completion.

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The Bigger Lesson

AI hallucination is not just about wrong answers.

It is about how we design intelligence.

If we reward confidence over truth, we get confident mistakes.

If we reward honesty over guessing, we get more reliable systems.

The future of AI will not only depend on smarter models.

It will depend on better evaluation.

Better incentives.

And better expectations from users.

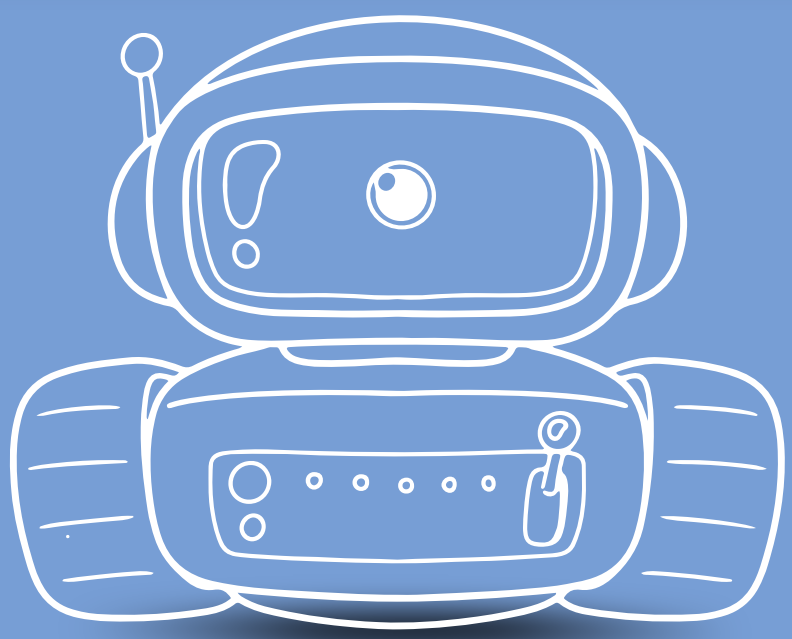
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Final Takeaway

AI models hallucinate because they are often trained and tested like overconfident exam takers.
When they don't know the answer, guessing can look better than admitting uncertainty.
That's the real problem.
AI should not just be trained to answer.
It should be trained to know when not to answer.
Because sometimes the smartest response is:
"I don't know."



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